Multi-Window Timeframe Temporal Features for Stroke-Risk Prediction

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2026 18th International Conference on Knowledge and Smart Technology (KST) | 979-8-3315-7082-8/26/$31.00 ©2026 IEEE | DOI: 10.1109/KST67832.2026.11431866

***Abstract*—Stroke is a major global health burden. The early stroke risk prediction is essential for reducing disability and mortality. Most existing machine-learning approaches rely on single-shot clinical records, which fail to capture the longitudinal health dynamics that precede stroke onset. To address this limitation, this study constructs temporal representations across three retrospective observation windows. Statistical, temporal, and cross-window features were generated to capture both short-term variability and long-term physiological trends, and three hierarchical feature sets were designed to evaluate the impact of increasing feature-engineering complexity. ANOVA F-tests and Principal Component Analysis (PCA) were used for dimensionality reduction. Model performance was subsequently evaluated using Logistic Regression with Leave-One-Out Cross-Validation (LOOCV) under conditions of severe class imbalance. The results showed that the temporal model outperformed the single-shot baseline with a maximum G-Mean of 0.638. McNemar's test further confirmed that temporal representations yielded statistically significant improvements and indicated minimal differences between PCA and ANOVA in prediction behavior. Overall, the results indicate that incorporating temporal information substantially enhances early stroke-risk prediction, providing more reliable decision-support capability than conventional single-time-point approaches.**

Keywords—component, stroke prediction, temporal features, single-shot data, machine learning, logistic regression, feature engineering, PCA, F-ANOVA

I. Introduction

Stroke remains a major global health concern, with more than 1.3 million new cases reported annually in Europe and one death occurring every four minutes worldwide. The condition frequently leads to long-term disability, substantial healthcare costs, and premature mortality. However, an estimated 40% of stroke cases could be prevented through early detection and timely intervention [1], [2].

Machine learning (ML) has shown promising potential for stroke-risk prediction. Prior studies have explored various ML models using demographic, clinical, and behavioral features. Wang et al. applied XGBoost and Logistic Regression with feature-importance–based feature selection for mortality prediction. Their results demonstrated that XGBoost outperformed Logistic Regression, achieving 98.1% accuracy and an AUC of 0.95, compared with 83.0% accuracy and an AUC of 0.77 [3]. Emon et al. compared ten ML classifiers and found that a weighted voting ensemble yielded the highest performance, achieving 97% accuracy with minimal false positives and negatives [4]. Das et al. evaluated nine algorithms, including support vector machines (SVM), K-nearest neighbor (KNN), XGBoost, AdaBoost, Random Forest, Decision Tree, LightGBM, and Logistic Regression, and reported Random Forest as the best-performing model with 98.4% accuracy [5]. Ensemble approaches have also been effective as reported in Dritsas and Trigka's study, which achieved a 98.9% AUC using a stacking-based framework [6]. More recently, Chowdhury et al. incorporated lag features and rolling statistical functions into an XGB-RS model enhanced with XAI techniques, achieving 98.06% accuracy and 99.12% specificity [7].

Despite these strong results, most existing models rely heavily on cross-sectional (single-shot) data, where each patient contributes only one clinical record. Such representations overlook longitudinal health dynamics—such as rising blood pressure, fluctuating lipid levels, or declining renal function—that often precede stroke onset. Consequently, single-shot models fail to leverage ML's ability to capture nonlinear and time-dependent patterns, limiting their clinical robustness.

In real-world healthcare systems, longitudinal electronic health records (EHR) offer richer temporal information but introduce additional challenges. Unlike curated research datasets, real-world EHR are irregular, incomplete, and heterogeneous. Patients visit clinics at inconsistent intervals, resulting in unaligned temporal records and uneven spacing between visits. Furthermore, key laboratory measurements, such as lipid profiles, glucose levels, and estimated glomerular filtration rate (eGFR), are not collected at every visit, leaving many observation windows partially or entirely incomplete. These issues complicate the construction of reliable temporal features and hinder the development of models suitable for clinical deployment.

According to these limitations, this study proposes a temporal-feature framework that organizes longitudinal EHR

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data into standardized multi-window representations to capture meaningful health trajectories. The proposed approach is benchmarked against traditional single-shot baselines to quantify the added predictive value of modeling temporal patterns. By addressing both the methodological gap and the practical challenges of real-world EHR, this work aims to advance stroke-risk prediction toward more clinically robust and temporally aware models.

II. Material and Methods

A. Data Collection

This study employed a real-world clinical dataset comprising 245,978 electronic health records collected from 19,473 patients at Chokchai Hospital, Thailand, between October 2014 and June 2024. Among these patients, 16,696 (85.7%) had no history of stroke events, while 2,777 (14.3%) experienced at least one stroke occurrence. Stroke events were identified using the International Classification of Diseases, 10th Revision (ICD-10), based on codes I60–I68 [8].

The original dataset contained 45 attributes. In this study, 16 clinically significant variables known to be associated with stroke risk were selected [3-10], including systolic blood pressure (BPS), diastolic blood pressure (BPD), high-density lipoprotein (HDL), total cholesterol, triglycerides, body mass index (BMI), eGFR, gender, age, atrial fibrillation (AF), alcohol consumption status, smoking status, creatinine, dispensed medications, principal diagnosis, and comorbidity diagnosis (ICD-10 codes).

B. Data Selection

In this study, patient data were selected using strict diagnostic, temporal, and completeness criteria to enable reliable multi-window temporal feature extraction. Stroke cases were identified using ICD-10 codes I60–I68, with the reference date defined as the first stroke event for stroke patients and the last clinical visit for non-stroke patients. All records after the reference date were removed to eliminate post-event leakage.

Given the irregular and incomplete nature of real-world EHR, a standardized temporal structure was imposed. Three retrospective windows were constructed for each patient (Fig. 1). For stroke patients, the FIRST, MID, and LAST windows corresponded to 21–9, 18–6, and 15–3 months before the stroke event; the same intervals were defined relative to the last visit for non-stroke patients. This framework ensured consistent temporal alignment across heterogeneous visit patterns and varying data availability.

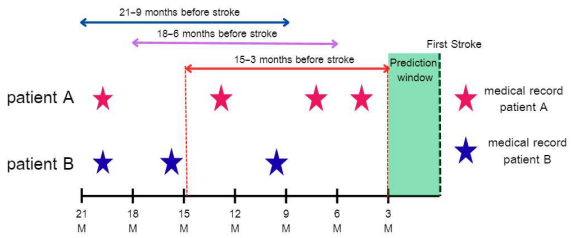

Fig. 1. Construction of three fixed retrospective temporal windows.

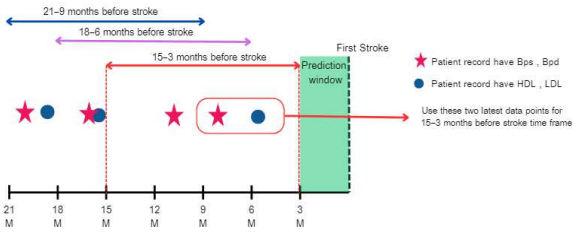

Fig. 2. Completeness constraint ensuring full clinical feature coverage within each temporal window.

A completeness constraint was then applied to ensure that temporal features could be computed uniformly across all individuals (Fig. 2). Each patient was required to satisfy two conditions:

1) each window (FIRST, MID, LAST) must contain at least one clinical visit

2) all core clinical variables required for temporal feature extraction must be present within every window, even if the window contained only a single visit.

This constraint minimized the impact of missing data and ensured that temporal representations were comparable across patients.

After applying all temporal alignment and completeness criteria, a total of 3,976 patients remained in the final dataset, consisting of 148 stroke cases and 3,828 non-stroke cases. The resulting cohort provides a temporally consistent and feature-complete foundation for extracting statistical and temporal representations used in downstream modeling.

C. Data Preprocessing

Data preprocessing was conducted to ensure clean and consistent clinical records before temporal segmentation. All personally identifiable information was removed, retaining only hospital numbers (HN) as anonymized identifiers. Data were harmonized by standardizing date formats, measurement units, and variable names, while implausible or duplicated entries were replaced with missing values. Binary clinical attributes (e.g., atrial fibrillation, smoking, alcohol use, heart disease, hypertension, diabetes, statin and antihypertensive use [9], [10]) were encoded in a {0,1} schema, and diagnostic fields were normalized using ICD-10 codes. These steps established a standardized and reliable dataset for constructing temporal windows and extracting features.

D. Feature Extraction

In this study, feature extraction was performed by combining clinically relevant variables, derived biochemical indices, statistical descriptors, and temporal dynamics obtained from three retrospective observation windows (FIRST, MID, and LAST). A derived cardiovascular biomarker, the total cholesterol–to–HDL ratio (TC:HDL) [11], was first computed following (1), as it is widely recognized as an indicator of lipid imbalance and stroke risk:

$$TC{:}HDL = Total\ Cholesterol/HDL \quad (1)$$

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For each of the ten numerical clinical variables (BPS, BPD, HDL, LDL, FBS, BMI, eGFR, creatinine, total cholesterol, triglycerides), six statistical descriptors—mean, minimum, maximum, standard deviation, first value, and last value—were computed across all measurements within the three windows. These features capture central tendency, variability, and short-term fluctuations beyond single-time-point representations.

To characterize long-term physiological trends, temporal descriptors were derived by comparing values between the FIRST and LAST windows, including delta change, slope of change, cross-window differences in minimum and maximum values, and differences in measurement counts. Delta change and slope were computed as

$$\Delta X = X_{LAST} - X_{FISRT} \tag{2}$$

$$Slope(X) = (X_{LAST} - X_{FISRT}) / (t_{LAST} - t_{FISRT}) \tag{3}$$

These temporal features capture longitudinal trends, such as increasing blood pressure or declining renal function, that cannot be observed from isolated clinical measurements. Demographic and categorical variables (age, sex, smoking, drinking, atrial fibrillation) were extracted from the most recent clinical record, together with six additional clinical risk factors: diabetes, hypertension, heart disease, statin use, antihypertensive use, and the TC:HDL ratio. All features were grouped into three hierarchical sets:

- Extract Set 1 (35 features): core demographic and clinical variables (10 numerical variables × 3 temporal windows + 5 categorical features).
- Extract Set 2 (115 features): extends Set 1 by adding detailed temporal and statistical descriptors for each numerical variable. This results in a total of 5 + (11 × 10) = 115 features, which consist of 5 demographic and categorial variables and 110 derived features generated by applying 11 temporal and statistical descriptors to 10 numeric clinical variables.
- Extract Set 3 (121 features): extends Set 2 by incorporating six additional clinical and biochemical risk factors (i.e., *diabetes, hypertension, heart disease, statin use, antihypertensive use, and the TC:HDL ratio). These additions result in* a feature set of 115 + 6 = 121 features.

A list of features is shown in Table I.

TABLE I. LIST OF FEATURES AND DESCRIPTIONS

Category	Feature	Type
Core Feature	Gender	Categorical
	Age	Numerical
	Atrial Fibrillation (AF)	Categorical
	Alcohol Drinking Status	Categorical
	Smoking Status	Categorical
	Systolic Blood Pressure (BPS)	Numerical
	Diastolic Blood Pressure (BPD)	Numerical
	High-Density Lipoprotein (HDL)	Numerical
	Low-Density Lipoprotein (LDL)	Numerical
	Fasting Blood Sugar (FBS)	Numerical
	Body Mass Index (BMI)	Numerical
	Estimated Glomerular Filtration Rate (eGFR)	Numerical
	Creatinine	Numerical
Other Feature	Dispensed drugs	Categorical
	Principal diagnosis	Categorical
	Comorbidity diagnosis	Categorical
Feature Extraction	Heart disease	Categorical
	Hypertension	Categorical
	Diabetes	Categorical
	Statin flag	Categorical
	Antihypertensive flag	Categorical
	TC:HDL ratio	Numerical

E. Feature Selection

Feature selection followed a two-step supervised procedure. First, the ANOVA F-test was used to rank numerical features by their discriminative power, assigning higher priority to features with lower p-values [12]. A forward top-*k* strategy was then applied, where features were added sequentially according to the ANOVA ranking (e.g., k = 5, 10, 15, …). Logistic Regression with class weighting was trained on each subset, and the optimal k was selected based on the highest G-Mean under LOOCV. This process was performed separately for Extract Set 1, Extract Set 2, and Extract Set 3 to assess how feature-engineering complexity affects supervised feature reduction.

F. Dimensionality reduction

Principal Component Analysis (PCA) was used to reduce redundancy among correlated temporal features and improve model generalization [13]. Numerical features were standardized prior to PCA, and the number of components k was varied incrementally (e.g., k = 5, 10, 15, …). Logistic Regression with class weighting was trained on each PCA-transformed set, and LOOCV with G-Mean determined the optimal k. PCA was applied independently to Extract Sets 1–3, allowing evaluation of its effectiveness across different feature complexities and enabling direct comparison with the supervised ANOVA-based approach.

G. Classification model

The stroke-risk prediction task was formulated as a binary classification problem. Because the dataset was highly imbalanced, Logistic Regression with class_weight = 'balanced' was used to reduce bias toward the majority class. Logistic Regression was selected for its interpretability, efficiency, and suitability for clinical decision support. Additionally, Leave-One-Out Cross-Validation (LOOCV), where each patient was held out once as the test case, was applied to ensure patient-level independence, prevent data leakage, and yield a reliable estimate of model generalization.

Three key performance metrics were used**:** sensitivity**,** specificity, and the geometric mean (G-Mean)**.** Sensitivity and specificity quantify the model's ability to correctly identify stroke and non-stroke patients, respectively, while G-Mean—a balance between the two—is defined as:

$$G\text{-}Mean = sqrt(Sensitivity \times Specificity) \tag{4}$$

G-Mean was emphasized because it offers a fairer assessment for highly imbalanced medical datasets than overall accuracy, ensuring that the model performs well for both minority (stroke) and majority (non-stroke) classes.

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Lastly, McNemar's test was implemented to evaluate two paired classifiers by examining disagreement cases, where one model predicts correctly while the other predicts incorrectly.

III. Result and Discussion

A. Comparison Between Single-Shot Baseline and Temporal

In this study, only the most recent clinical record was used for single-shot baseline. We compared the proposed temporal model constructed from three retrospective windows with the single-shot baseline for the first analysis. Unlike the single-shot approach, the temporal representation captures longitudinal health patterns such as biomarker trends and gradual physiological changes. As shown in Table II, the temporal model (Extract Set 1 – LAST) achieved a higher G-Mean (0.627) than the single-shot baseline (0.611). Although the difference is small, McNemar's test was applied to determine whether the improvement is statistically meaningful rather than occurring by chance.

McNemar's test ($\chi^2 = 50.70$, $p = 7.50 \times 10^{-13}$) further supports the superiority of the temporal model. As shown in Fig. 3, the temporal model correctly predicted considerably more cases in which the baseline model failed (394 vs. 217). This large discrepancy indicates a statistically significant improvement ($p < 0.001$) achieved by the temporal model.

These findings confirm that the observed performance gain is unlikely to be due to chance. Incorporating temporal information therefore provides meaningful predictive value beyond single-shot clinical records, resulting in more reliable stroke-risk predictions.

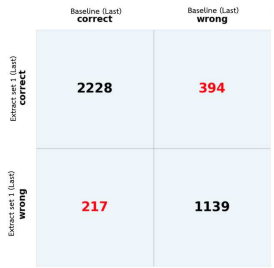

Fig. 3. McNemar comparison between temporal features (Extract Set 1 – LAST) and single-shot baseline (Baseline – LAST).

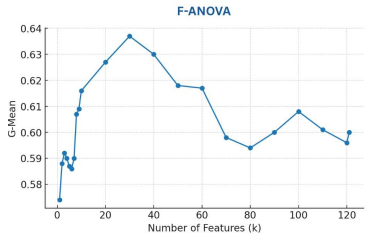

Fig. 4. G-Mean performance of ANOVA across different numbers of features (k).

TABLE II. OVERALL PERFORMANCE COMPARISON ACROSS FEATURE SETS AND DIMENSIONALITY REDUCTION METHODS

Feature Extraction	Feature Set	Method		Number of feature / components	Sensitivity	Specificity	G-Mean
Single shot	Base line (15)	LAST	No reduce	15	0.5883	0.6362	0.6117
			F-ANOVA	6	0.6084	0.6413	0.6241
			PCA	9	0.6152	0.6431	0.6287
			F-ANOVA+PCA	3	0.6251	0.6197	0.6223
Temporal feature	Extract Set 1 (35)	MEAN	No reduce	35	0.5873	0.6512	0.6183
			F-ANOVA	30	0.6224	0.6423	0.6322
			PCA	35	0.5942	0.6623	0.6274
			F-ANOVA+PCA	15	0.6283	0.6279	0.6281
		LAST	No reduce	35	0.5943	0.6613	0.6273
			F-ANOVA	30	0.5954	0.6631	0.6284
			PCA	30	0.6013	0.6614	0.6307
			F-ANOVA+PCA	29	0.5878	0.6509	0.6192
	Extract Set 2 (115)	MEAN	No reduce	115	0.5274	0.6703	0.5943
			F-ANOVA	4	0.5813	0.6364	0.6081
			PCA	30	0.6014	0.6203	0.6107
			F-ANOVA+PCA	2	0.6553	0.5362	0.5928
		LAST	No reduce	115	0.5332	0.6743	0.5997
			F-ANOVA	20	0.6154	0.6334	0.6241
			PCA	**30**	**0.6153**	**0.6624**	**0.6385**
			F-ANOVA+PCA	19	0.6216	0.6483	0.6348
	Extract Set 3 (121)	MEAN	No reduce	121	0.5063	0.6842	0.5884
			F-ANOVA	40	0.6084	0.6453	0.6263
			PCA	30	0.5743	0.6493	0.6107
			F-ANOVA+PCA	12	0.6418	0.6214	0.6315
		LAST	No reduce	121	0.5273	0.6842	0.6007
			F-ANOVA	**30**	**0.6283**	**0.6454**	**0.6367**
			PCA	50	0.5883	0.6703	0.6279
			F-ANOVA+PCA	**22**	**0.6351**	**0.6415**	**0.6383**

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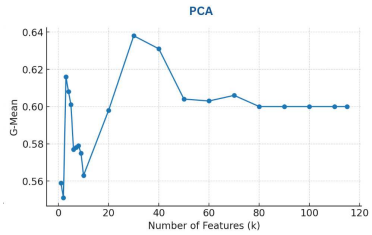

Fig. 5. G-Mean performance of PCA across different numbers of components (k).

B. Performance of Feature selection (F-ANOVA)

The dimensionality-reduction analysis first evaluated supervised feature selection using the ANOVA F-test. Features were ranked by their F-scores, and the top-k features were added incrementally to assess performance changes. As shown in Table II, model performance improved as more informative features were included, reaching its peak with the Extract Set 3 – LAST configuration (G-Mean = 0.637 at approximately 30 features, Fig. 4). However, adding more features beyond this point provided no further benefit and, in some cases, reduced model stability.

C. Performance of PCA

Next, PCA was evaluated as an unsupervised dimensionality-reduction method. PCA projects the original feature space into a set of orthogonal components ordered by the amount of variance explained. The number of components was gradually increased (k = 1 to K) to analyze the trade-off between dimensionality and predictive performance.

As presented in Table II, PCA achieved its highest performance using the Extract Set 2 – LAST feature set with approximately 30 components, yielding a G-Mean of 0.638. This trend closely aligns with the pattern observed from ANOVA. PCA was also effective in mitigating multicollinearity and compressing highly correlated variables into compact latent representations (Fig. 5).Comparison Between Single-Shot and F-ANOVA + PCA Models

To evaluate the benefit of temporal information and dimensionality reduction, we compared the proposed F-ANOVA + PCA model with the traditional single-shot baseline using Extract Set 3 (LAST). The F-ANOVA + PCA model (top 30 ANOVA features followed by PCA) achieved G-Mean = 0.6383, outperforming the single-shot model, which obtained G-Mean = 0.611.

McNemar's test (Fig. 6) further confirmed a significant difference between the two models: the single-shot model correctly classified 223 cases misclassified by the F-ANOVA + PCA model, whereas the proposed model corrected 286 cases missed by the single-shot approach (χ^2 = 7.55, p = 0.006). These findings indicate that incorporating temporal windows together with ANOVA to PCA dimensionality reduction provides a notably stronger predictive capability than relying on a single recent measurement.

D. Comparative with Previoues Studies

Previous studies have reported high predictive performance using demographic, behavioral, and limited clinical information, as shown in Table III. However, most existing models rely on single-shot data, where only one record per patient is used, restricting their ability to capture longitudinal health dynamics such as progressive hypertension, renal decline, or lipid fluctuations.

Many works, including those by Wang et al., Emon et al., and Das et al., primarily used demographic or self-reported attributes, with minimal incorporation of objective laboratory data. Although Chowdhury et al. introduced some temporal features, their approach did not employ a structured multi-window EHR design or comprehensive physiological tracking.

In contrast, our method uses real-world longitudinal EHR with integrated temporal features, demographic factors, and laboratory measurements. This provides a more accurate representation of physiological changes over time and a stronger foundation for clinical decision-making. While the raw accuracy is lower due to the complexity of real hospital data, the proposed framework fills a key methodological gap by modeling true temporal health trajectories essential for early stroke-risk prediction

Fig. 6. McNemar comparison between temporal features (Extract Set 1 – LAST) and single-shot baseline (Baseline – LAST).

TABLE III. COMPARISON OF DATA MODALITIES AND PREDICTIVE PERFORMANCE ACROSS PRIOR STUDIES

	Single-shot	Temporal features	Demographic Info	Laboratory Measurements	Accuracy	Sensitivity	Specificity
Ours (F-ANOVA + PCA)		/		/	0.641	0.635	0.641
Wang et al. [3]	/		/		0.958	0.984	0.933
Emon et al [4]	/		/		0.970	1.000	0.900
Das et al. [5]	/		/		0.984	0.983	-
Dritsas and Trigka [6]	/		/		0.989	0.974	-
Chowdhury et al. [7]	/		/		0.980	0.884	0.991

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IV. Conclusions

This study presented a temporal-feature framework for stroke-risk prediction using real-world longitudinal EHR, overcoming key limitations of single-shot models. By structuring patient records into three standardized retrospective windows and applying completeness criteria, the method captures meaningful physiological trends that cannot be observed from single-timepoint data.

Our results show that temporal features markedly improve predictive performance over the single-shot baseline. The combination of F-ANOVA and PCA further enhances model accuracy, achieving the highest G-Mean test confirms that these improvements are statistically significant. Overall, the findings emphasize the value of temporal health trajectories and dimensionality reduction in modeling complex clinical outcomes. The proposed framework establishes a foundation for more robust and clinically applicable stroke-risk stratification using routine hospital EHR

Acknowledgment

This study is financially supported by Faculty of Informatics, Burapha University. This work was supported by Suranaree University of Technology (SUT), Thailand Science Research and Innovation (TSRI), and National Science Research and Innovation Fund (NSRF) (NRIIS no.195615).

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